Arab Open University - Egypt



Faculty of Artificial Intelligence

Information Technology and Computing Department

AI Personalized Nutrition
Recommendation System

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Declaration

I hereby certify that this material, which I now submit for assessment on the program of study leading to the award of bachelor of Information Technology and Computing in Computer Science is entirely my own work, that I have exercised reasonable care to ensure that the work is original, and does not to the best of my knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of my work.

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Mario Fayez

Abstract

Text here.

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Abbreviations

AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
BMI	Body Mass Index
BMR	Basal Metabolic Rate
CSV	Comma-Separated Values
DASH	Dietary Approaches to Stop Hypertension
DFD	Data Flow Diagram
ERD	Entity-Relationship Diagram
HMAC	Hash-based Message Authentication Code
KNN	K-Nearest Neighbours
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
PBKDF2	Password-Based Key Derivation Function 2
PDF	Portable Document Format
PIN	Personal Identification Number
SQL	Structured Query Language
SR Legacy	Standard Reference Legacy (USDA food data release)
TDEE	Total Daily Energy Expenditure
UI	User Interface
USDA	United States Department of Agriculture

Chapter 1
Introduction

Start with Background …….

1.1 Problem definition

Write the problems……. Add some references

1.2 Significance of the Problem

The importance of the problem and its effects ……

1.3 Project Objectives

The project objectives can be split into four main points as follows:

1. ..
2. ..
3. ..
4. ..

1.4 Scope of the Project

1.5 Project Methodology

…… as shown in Fig.1-1.

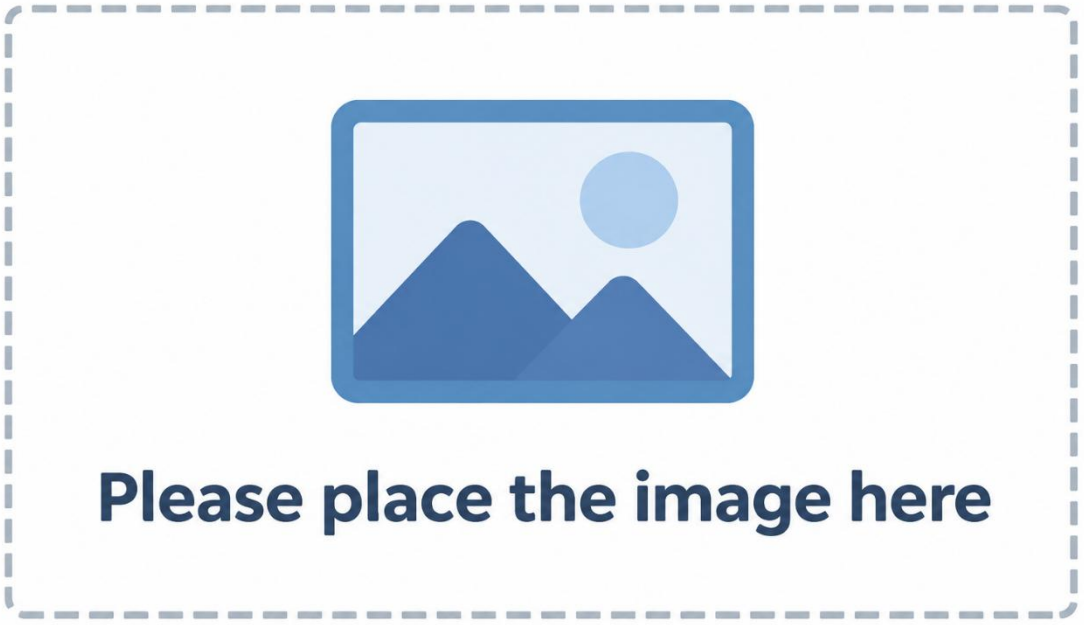


Figure 1-1 Write the figure caption here……

1.6 Structure of the Report

For Example:

Chapter 2 presents ……..

Chapter 3 focuses on ……..

Chapter 4 describes ……..

Chapter 5 discusses ……...

Finally, Chapter 6 concludes the report by summarizing the main findings, reflecting on the overall outcomes of the project, and emphasizing its contributions.

Chapter 2
Literature Review

Introduction to this chapter here……. (at least **2000** words for this chapter)

2.1 Background

2.1.1 Subsection

Text text ….. as shown in Fig. 2-1.

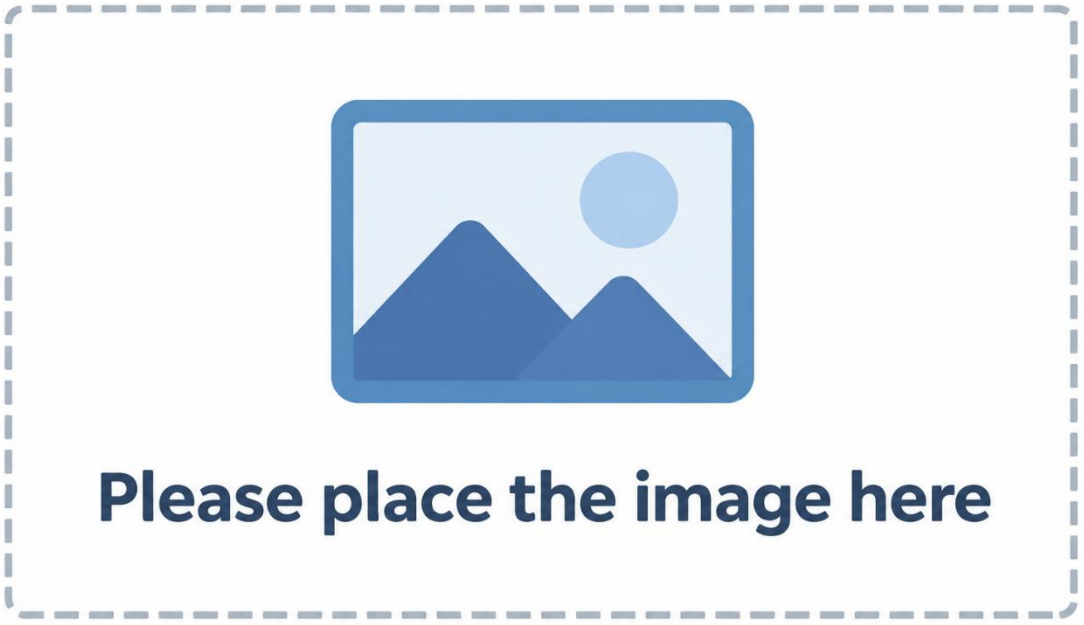


Figure 2-1 Write the figure caption here……

2.1.1.1 Sub-Subsection

Text text ……

2.1.1.2 Sub-Subsection

Text text …… as shown in Fig. 2-2.

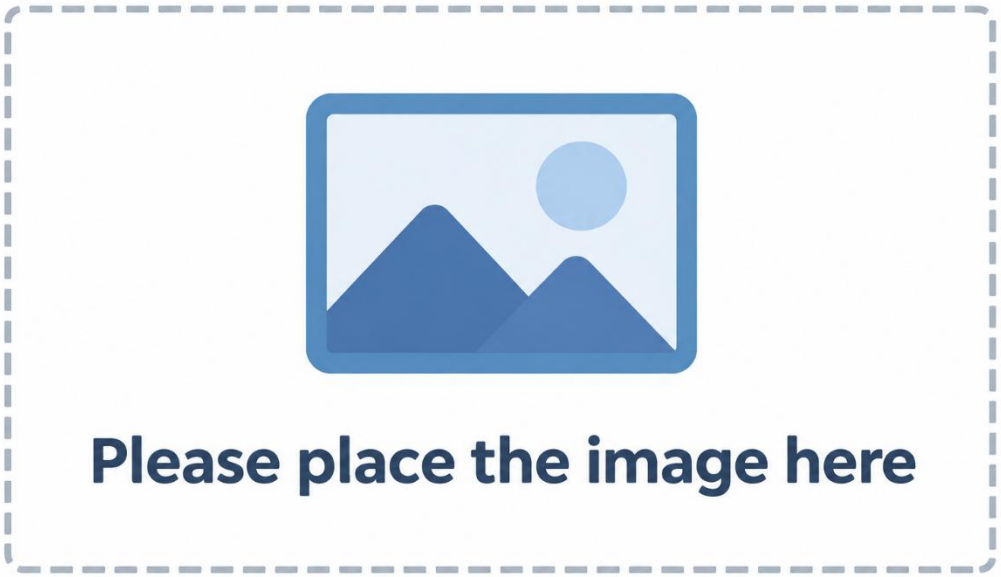


Figure 2-2 Write the figure caption here……

2.1.2 Subsection

Transfer learning refers …….

2.1.2.1 Sub-subsection

Text text ……

2.1.2.2 Sub-subsection

Text text ……

2.2 Related Projects

Text text……

2.2.1 Project 1: Name

Text text……

2.2.2 Project 2: Name

Text text……

2.2.3 Project 3: Name

Text text……

2.3 Related Research

Test text …….

2.4 Summary

In this part, a summary is provided …….

..

Chapter 3
Requirements and Analysis

This chapter provides a detailed description of the system requirements and presents the analysis of the problem.....

3.1 Functional Requirements

Write the text here…….

Functional requirements define what the system should do……. User Registration and Login, Recommendation, Search Functionality, Admin Dashboard, Classification, ….

You can write tabled as shown: Table 3-1 summarizes the methods ……..

Table 3-1 Write the table title here……

Table Head	Table Head	Table Head
Text	2025	Text
Text	2026	Text
Text	2026	Text

3.2 Non-Functional Requirements

Write the text here…….

Non-functional requirements describe the quality attributes of the system…… Performance, Usability, Security, Reliability, Scalability, Portability, ……..

3.3 System Analysis

Write the text here…….

3.3.1 Software Requirements

Write the text here…….

3.3.2 Hardware Requirements

Write the text here…….

3.3.3 Other Requirements

Write the text here…….

3.4 System Modeling

Write the text here…….

3.4.1 Use Case Diagram

Write the text here…….

3.4.2 Data Flow Diagram

Write the text here…….

3.4.3 Entity–Relationship Diagram

Write the text here…….

..

..

3.5 Testing and Evaluation Plan

Write the text here…….

Testing will be conducted to ensure that all system components function as intended.

Evaluation metrics …

3.6 Summary

Write the text here……..

Chapter 4
Design, Implementation, and Testing

Start with Introduction to this chapter……

4.1 Detailed Design

…… is illustrated in Fig. 4-1. First, …….. Second, ……., ……., …….

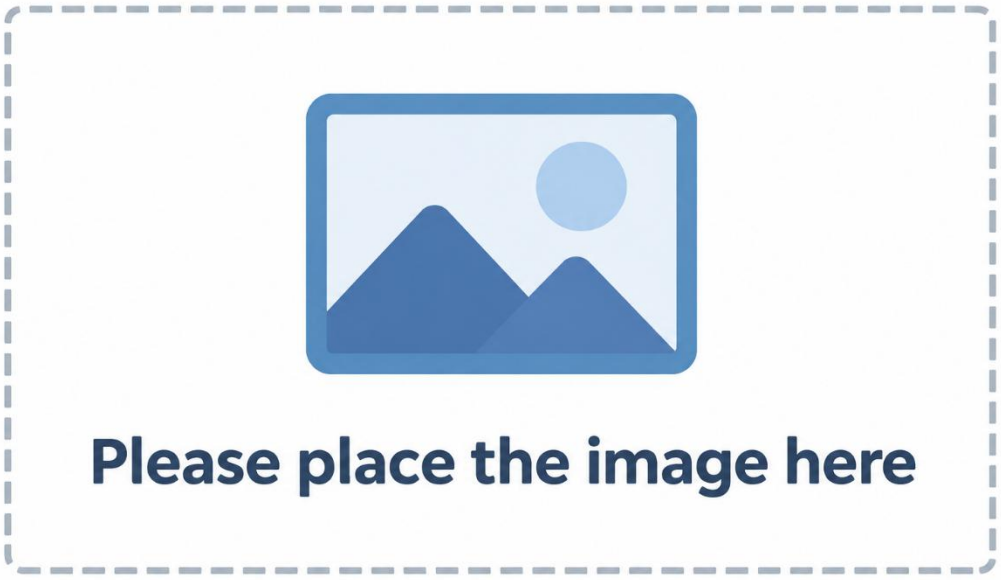


Figure 4-1 Write the figure caption here……

4.2 Implementation

Write the text here……..

4.2.1 Experimental Setup

Write the text here……..

4.2.2 Description of Datasets

Three benchmark datasets for …… were used: 1)……..., 2)……., and 3)……...

4.2.3 User Interface ……

4.2.4 Application ……

Write the text here……..

Example for Table:

…….. is shown in Table 4-1.

Table 4-1 Write the table title here……

Table Head	Table Head
Text	Text
Text	Text
Text	Text

4.3 Testing

Write the text here……..

Unit test, Usability test, Security test, ….

4.4 Summary

Write the text here……..

Chapter 5
Results and Discussion

Start with Introduction to this chapter……

5.1 Findings

Present all the results (products, experimental findings, theories, etc.) generated during the project…

5.1.1 Experimental Results

Presents the results…….

5.1.2 Ablation Study

Test each part of your project……

5.2 Goals Achieved

Describes the degree to which the findings support the original objectives laid out for the project. The goals may be partially or fully achieved or exceeded…

5.3 Further Work

Describes two things: firstly, new areas of an investigation prompted by developments in this project, and secondly parts of the current work which were not completed due to time constraints and/or problems encountered…

5.3.1 Enhancements and Future Features

Write the text here……..

5.3.2 Limitations and Incomplete Work

Write the text here……..

5.4 Social and Ethical Implications

Write the text here…….

Provide clear evidence for the consideration of social, professional, legal and ethical issues in relation to the project and it should be explained in one or more paragraphs.

5.5 Summary

Write the text here…….

Chapter 6
Conclusions

Start with Introduction to this chapter……

6.1 Summary of the Work

Write the text here……..

6.2 Main Contributions

Write the text here……...

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